

GANGNEUNG, Korea, Republic of

WMO: 471050

Lat: 37.7514N

Lon: 128.8911E

Elev: 27

StdP: 101.00

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-5.9	-24.7	0.4	-5.2	-22.7	0.5	-3.9	9.5	-1.4	8.4	-1.1	5.2	230	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	33.4	23.8	31.9	23.2	30.5	22.6	25.5	30.3	24.7	29.3	24.1	28.6	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.1	27.8	23.4	18.3	27.2	22.7	17.5	26.7	78.5	30.4	75.5	29.4	72.9	28.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.6	5.9	-11.1	35.4	2.6	1.3	-13.0	36.3	-14.5	37.1	-15.9	37.8	-17.8	38.7		
(5)			-12.8	26.3	2.2	1.0	-14.4	27.0	-15.7	27.6	-16.9	28.2	-18.5	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.1	2.8	7.1	13.1	18.0	21.3	25.0	25.3	20.7	16.0	9.7	3.5		
	DBStd	9.10	3.53	3.62	4.20	4.36	3.96	3.25	3.71	3.03	2.62	2.97	3.89	3.91		
	HDD10.0	867	277	203	112	17	0	0	0	0	0	2	53	204		
	HDD18.3	2358	535	435	350	167	56	8	1	0	6	82	259	461		
	CDD10.0	2213	0	1	21	109	249	338	464	476	321	188	43	2		
	CDD18.3	661	0	0	1	9	46	96	206	217	77	9	1	0		
	CDH23.3	5231	0	0	1	56	322	653	1929	1907	341	22	0	0		
	CDH26.7	1798	0	0	0	9	73	179	790	678	68	2	0	0		
(14) Wind	WSAvg	2.6	3.4	3.1	3.0	2.9	2.5	1.9	2.0	1.9	2.1	2.5	2.9	3.3		
(15) Precipitation	PrecAvg	1403	52	54	70	79	79	111	231	277	220	107	80	39		
	PrecMax	2096	248	196	174	248	277	343	758	1137	559	348	348	234		
	PrecMin	924	0	0	8	1	3	14	58	55	0	2	0	0		
	PrecStd	279	50	48	43	56	55	70	134	178	149	90	75	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	15.2	21.8	27.9	31.2	32.9	35.2	35.0	31.2	25.4	21.6	14.8		
		MCWB	5.7	7.8	10.8	14.8	17.0	20.6	24.5	25.1	23.2	15.4	13.3	8.3		
	2%	DB	9.1	12.2	18.7	24.8	28.7	30.6	33.7	33.3	28.6	23.5	19.3	12.4		
		MCWB	3.6	5.6	9.4	13.4	15.8	19.3	24.1	24.3	21.0	15.8	11.7	6.9		
	5%	DB	7.5	10.1	16.3	22.7	26.8	28.7	32.2	31.8	26.7	22.2	17.4	10.5		
		MCWB	2.4	4.4	8.2	12.4	15.4	19.1	23.6	24.2	20.4	15.4	10.9	5.3		
	10%	DB	6.1	8.3	14.2	20.5	24.9	27.1	30.7	30.2	25.1	20.8	15.7	8.9		
		MCWB	1.5	3.3	6.9	11.2	15.0	18.8	23.2	23.5	19.7	15.0	10.1	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	8.7	12.2	17.4	19.5	23.1	26.4	26.7	23.9	19.2	15.0	9.8		
		MCDB	10.2	13.7	19.9	24.6	26.0	27.6	32.2	31.6	28.7	21.7	18.8	14.0		
	2%	WB	4.6	6.4	10.4	14.8	18.0	21.9	25.4	25.7	22.7	18.0	13.4	7.7		
		MCDB	7.6	10.8	17.1	21.9	25.2	26.5	30.7	30.5	26.6	20.9	17.2	11.6		
	5%	WB	3.3	5.1	8.9	13.5	17.0	20.9	24.6	25.1	21.7	17.1	12.1	5.9		
		MCDB	6.4	9.1	14.5	20.2	23.3	25.7	29.9	29.3	25.1	20.3	16.0	9.4		
	10%	WB	2.2	3.8	7.6	12.3	16.1	20.1	24.0	24.5	20.7	16.1	10.8	4.7		
		MCDB	5.2	7.4	13.2	18.5	22.3	24.8	29.0	28.6	23.9	19.7	14.5	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.3	8.0	8.6	8.2	6.8	5.6	5.9	6.6	7.7	7.3	7.1		
		MCDBR	7.8	9.1	10.5	11.1	11.0	9.8	7.6	7.9	8.5	9.0	8.9	8.1		
	5% WB	MCWBR	5.4	5.8	5.9	5.5	4.7	3.9	2.9	3.1	4.2	5.0	5.5	5.7		
		MCWBR	6.9	8.5	9.8	10.0	9.2	7.8	6.8	6.7	7.0	7.0	7.5	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.425	0.491	0.590	0.632	0.637	0.688	0.648	0.575	0.501	0.494	0.457	0.415		
		taud	1.952	1.825	1.666	1.635	1.668	1.629	1.771	1.945	2.072	1.963	1.984	2.007		
	Ebn at Noon		706	706	676	681	689	654	677	718	749	698	668	684		
		Edh at Noon	137	178	229	250	246	255	220	181	151	153	133	122		
(44) All-Sky Solar Radiation	RadAvg		2.40	3.18	4.11	4.90	5.44	5.16	4.29	4.27	3.90	3.40	2.41	2.15		
	RadStd		0.21	0.30	0.41	0.38	0.53	0.43	0.60	0.54	0.47	0.29	0.31	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-2.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	-0.94	N/A	N/A	N/A	N/A	N/A	+97	+52

Nomenclature: See separate page